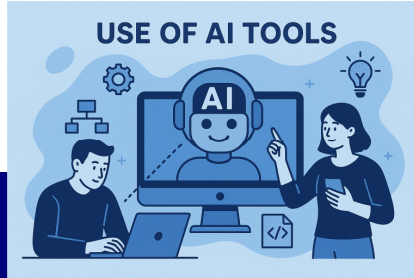


USE OF AI TOOLS



CS 59300

Application of Deep Learning Use of AI Tools

Instructor: Dr. Zesheng Chen

Prepared on August 28, 2025

ChatGPT 5 Version

PURDUE
UNIVERSITY
FORT WAYNE

HUNG-YI LEE (李宏毅)

Acknowledgement

HOME HONOR RESEARCH TALK PUE



INTRODUCTION TO GENERATIVE AI 2024 SPRING

<https://speech.ee.ntu.edu.tw/~hylee/genai/2024-spring.php>

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AI Tools

- ChatGPT 5
- Google NotebookLM

3

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Artificial Intelligence (Objective)

Machine Learning (Methods)

Deep Learning (Important Methods)

Generative Artificial Intelligence

Language Model

ChatGPT

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What Can ChatGPT 5 Do?

- Can write
- Can search
- Can code
- Can research
- Can draw
- Can help
- ...

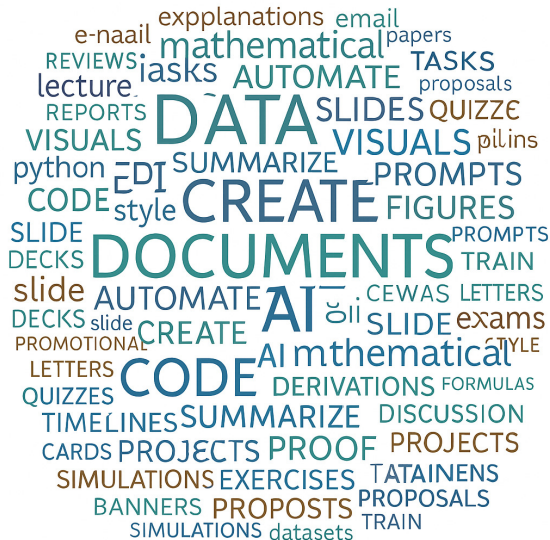
5

Communications Between Me and ChatGPT 5

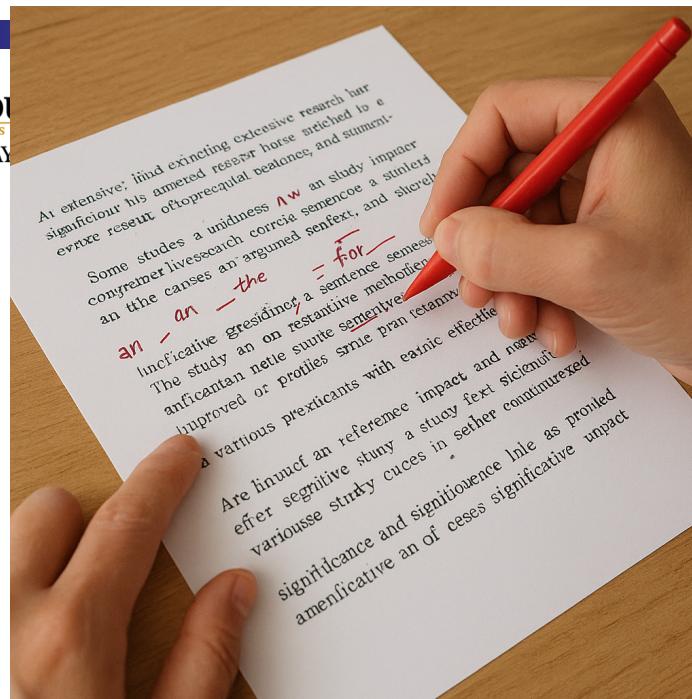
- <https://chatgpt.com/share/68b6f570-e184-800f-95b7-70faba01ad93>

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What Can ChatGPT 5 Do?



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Who is Zesheng Chen?

Could you please clarify which Zesheng Chen you're referring to? For example:

- Are you looking for information on an academic (e.g., a professor)?
- Are you referring to someone in industry or a public figure?

Any additional details (such as affiliation, field, or context) would help narrow down the research.

look for information on an academic

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Please list what you can do, at least 30 items, and for each item be concise. Then put what you can do into word cloud.

What is python code for creating the visual word cloud?

Thought for 45s

Here's a clean, runnable example that makes a word cloud from the capabilities list (keeps multi-word phrases intact) and saves it as `word_cloud.png`.

```
python
# pip install wordcloud matplotlib
from wordcloud import WordCloud
import matplotlib.pyplot as plt

# 1) Put your phrases here (use underscores to keep phrases together)
phrases = [
    "revise_papers", "recommendation_letters", "grant_proposals", "promotion_statements",
    "student_emails", "lecture_notes", "document_summaries", "apa_ieee_acm_formatting",
    "paper_search", "model_comparisons", "math_derivations", "latex_figures",
    "data_analysis", "data_visualization", "code_review",
    "slide_decks", "quizzes_exams", "case_studies", "coding_assignments",
    "concept_explanations",
    "python_java_c_web_code", "explain_ai_ml", "ml_workflows", "rl_adversarial_sims",
    "task_automation",
    "images_diagrams", "word_clouds", "timelines_concept_maps", "posters_banners",
    "workshop_content"
]
```

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Zesheng Chen

Education

Zesheng Chen received his B.E. (1998) and M.E. (2001) in Electronic Engineering from Shanghai Jiao Tong University, China [users.pfw.edu](#). He later obtained an M.S. (2005) and Ph.D. (2007) from the School of Electrical and Computer Engineering at the Georgia Institute of Technology [users.pfw.edu](#), where his research focused on computer network security and self-propagating worms [pfw.edu](#).

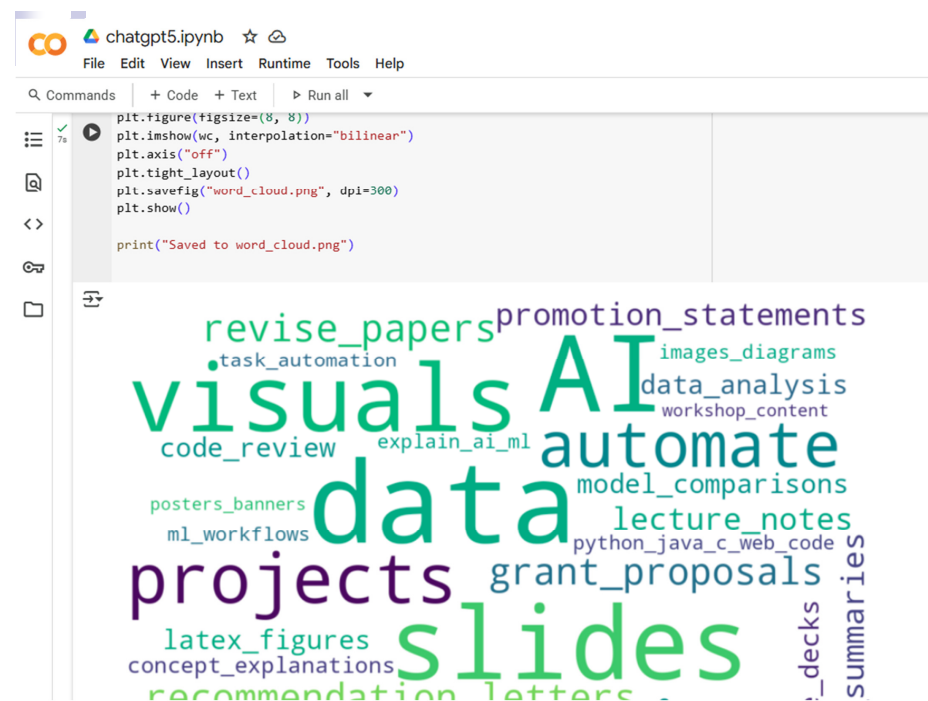
Academic Positions and Affiliations

Dr. Chen began his academic career as an Assistant Professor in Electrical and Computer Engineering at Florida International University (2007–2009) [users.pfw.edu](#). In 2009, he moved to Fort Wayne, Indiana, serving as a limited-term lecturer in the engineering and computer science departments at Indiana University–Purdue University Fort Wayne (IPFW) from 2009 to 2015 [users.pfw.edu](#). Since 2015, he has been on the faculty at Purdue University Fort Wayne, where he is currently an Associate Professor of Computer Science [pfw.edu](#). (During 2012–2015, he also gained industry experience as a software engineer at TransWorks [users.pfw.edu](#) before returning to academia.)

Research Areas and Interests

Dr. Chen's research interests span several areas in computer science and security, including:

- Machine Learning Security:** Adversarial attacks and defenses in speaker verification systems and network intrusion detection systems [users.pfw.edu](#).
- Network Security:** Modeling and defending against Internet worm and botnet attacks [users.pfw.edu](#).
- Internet of Things (IoT):** Security applications in IoT and smart devices [pfw.edu](#).



You
what is 533215 multiplied by 251349?

ChatGPT
533,215 multiplied by 251,349 equals 133,975,948,135.

🔊 📄 ↺ 🗑

what is 533215 multiplied by 251349?

Analysis

```
python
# Performing the multiplication
result = 533215 * 251349
result

Result
134023057035
```

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what is 533215 multiplied by 251349?

Let's calculate 533215×251349 **step by step** to avoid mistakes:

Step 1. Break numbers down

$$533215 \times 251349 = 533215 \times (251000 + 349)$$

Step 2. Multiply 533215×251000

Step 4. Add results

$$533215 \times 251349 = 133,836,965,000 + 186,091,035 \\ = \mathbf{134,023,056,035}$$

Final Answer:

`\boxed{134,023,056,035}` ✓ --- If you want, I can give you a ****short Python snippet**** to verify this calculation automatically. Do you want me to?

short Python snippet to verify the calculation

Here's a short Python snippet to **verify the calculation**:

```
python
# Simple multiplication check
a = 533215
b = 251349

print("Result:", a * b)
```

Copy code

Output:

```
makefile

Result: 134023057035
```

Copy code

This confirms that:

$$533,215 \times 251,349 = 134,023,057,035 \quad \checkmark$$

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find related works in adversarial attacks, detection, and defenses in network intrusion detection systems

To find the most relevant related works, could you please clarify:

1. Are you looking for recent academic papers (e.g., 2020–2025)?
2. Should the focus be only on deep learning-based NIDS (Network Intrusion Detection Systems), or also include classical approaches?
3. Do you want works covering all three: adversarial attacks, detection of adversarial activity, and defenses?

This will help me tailor the research to your needs.

1. Recent academic papers. 2. Deep learning-based NIDS. 3. All three.

Thanks! I'll gather recent academic papers (from 2020 to 2025) focused on adversarial attacks, detection, and defenses specifically in deep learning-based network intrusion detection systems (NIDS). This will include papers that explore how NIDS are attacked, how adversarial examples can be detected, and the strategies used to defend against them.

I'll get back to you shortly with a structured summary and relevant citations.

Searched for adversarial network...Looking for sources...

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+ Please summarize the main idea in the attached paper



Summary of the Paper

Title: *Ensemble Adversarial Defenses and Attacks in Speaker Verification Systems*

Authors: Zesheng Chen, Jack Li, Chao Chen

1. Problem Motivation

Speaker Verification (SV) systems are widely used in online banking, IoT, and voice-based authentication.

- **Vulnerability:** These systems rely on machine learning, making them **susceptible to adversarial attacks** — small, imperceptible perturbations in audio can cause misclassification.
- **Challenge:** No single defense method can effectively counter all types of adversarial attacks.

2. Proposed Approach

The paper introduces a **novel ensemble defense framework**, MEH-FEST-NA, which integrates two complementary strategies:

- **MEH-FEST:** Detects adversarial audio by monitoring **high-frequency energy** during short periods of silence.
- **Noise Adding (NA):** Adds random Gaussian noise to inputs to reduce the effect of perturbations.

Two variants of the ensemble system are proposed:

- **N-MEH-FEST-NA (Naive):** Sequentially applies MEH-FEST detection, then NA defense.
- **F-MEH-FEST-NA (Fine-tuned):** Dynamically decides whether to apply NA based on an energy threshold

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Create a picture for the course of "Application of Deep Learning"



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Create a short video for the course of "Application of Deep Learning"

I can definitely help you create a **short promotional video** for your course "Application of Deep Learning."

Since we want a smooth animation with visuals, we'll need to:

1. **Prepare a short script** (title, highlights, and key features).
2. **Design slides / frames** showing the content.
3. **Generate the video** with transitions, background music, and optional captions.

Before I proceed, could you confirm:

- Approximate **duration** you want (e.g., 30 sec, 60 sec)?
- Do you want **voice-over narration** or just **background music + text**?
- Preferred **style**: academic, modern tech, or creative?

With these details, I can generate a polished MP4 video for you.

Duration: 30 sec. Voice-over narration + background music. Academic style.

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Create the MP4 video now

Currently, I **cannot directly generate MP4 videos with synchronized voice-over and background music** in one step within this environment. However, I can guide you through a **complete workflow** to create your 30-second "Application of Deep Learning" promo video professionally — using **Python** or **free AI tools**.

I'll give you **two options**:

Option 1 — Python Script (MoviePy + gTTS)

This solution creates an **MP4 video** with:

Option 2 — Faster & Higher-Quality AI Tools (Recommended)

If you'd like a **professional academic-style animated video**, I recommend:

1. **Canva** → <https://www.canva.com/>
 - Import your course title + icons.
 - Add smooth animations & background music.
 - Export as MP4.
2. **Pictory AI** → <https://pictory.ai/>
 - Paste your script → auto-generates an animated AI video.
 - Choose **voice-over style** & **background music**.
 - Export in 1080p MP4.
3. **Synthesia** → <https://www.synthesia.io/>
 - Best for **AI-generated presenters** with narration.
 - Professional look for academic courses.

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Quiz

Help say "Hello World!" for 100 times!

Sure 😊 Here's a short Python script that prints "Hello World!" exactly 100 times:

python Copy code

```
for i in range(100):
    print("Hello World!")
```

Or, if you also want to **number** each line:

python Copy code

```
for i in range(1, 101):
    print(f"{i}: Hello World!")
```

Do you want me to make it even **fancier**, like printing each "Hello World!" with a **delay and colors**?

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Tips for Using ChatGPT

- Treat ChatGPT as an online **new** assistant, which has the ability of a normal person and does **not** know about you!
- Provide the **sufficient** information to ChatGPT.
- **"Magic words"**:
 - ☐ "let's think step by step!!!"
 - ☐ "answer by starting with "Analysis:""
 - ☐ "This is very important to my career."
 - ☐ "Let's work this out in a step by step way to be sure we have the right answer."
 - ☐ "Take a deep breath and work on this problem step-by-step."

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More Tips

- Paper "Principled Instructions Are All You Need for Questioning LLaMA-1/2, GPT-3.5/4"
- <https://arxiv.org/abs/2312.16171>
- <https://github.com/VILA-Lab/ATLAS>
- Findings:
 - ☐ No need to be polite with LLM so there is no need to add phrases like "please", "if you don't mind", "thank you", "I would like to", etc.,
 - ☐ Employ affirmative directives such as 'do,' while steering clear of negative language like 'don't'.
 - ☐ Add "I'm going to tip \$xxx for a better solution!"
 - ☐ Incorporate the following phrases: "You will be penalized"
 - ☐ Add to your prompt the following phrase "Ensure that your answer is unbiased and avoids relying on stereotypes."
 - ☐

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How GPT was Built?

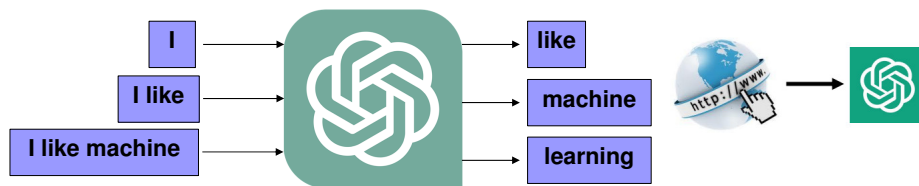
- Three Steps
 - ☐ Self-supervised learning from Internet
 - ☐ Supervised learning based on instruction fine-tuning
 - ☐ Reinforcement Learning from Human Feedback

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Self-supervised Learning from Internet

- Text solitaire



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Supervised Learning Based on Instruction Fine-tuning

- Provide correct answers to questions to train GPT
- Do NOT require a large number of training datasets

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			

Instruct GPT

<https://arxiv.org/abs/2203.02155>

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Reinforcement Learning from Human Feedback

- Improve the performance by learning from the human feedback



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AI Tools

- ChatGPT 5
- **Google NotebookLM**

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Google NotebookLM

■ <https://notebooklm.google/>

■ Functionalities

- ☐ Summarize key points
- ☐ Generate audio as two-people discussions
- ☐ Generate video presentation
- ☐ Generate mind map



Q & A

Have fun!!!